

SAMARTH KUMBLE

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PROFESSIONAL SUMMARY

Computer Science undergraduate (2027 batch) with hands-on experience across the full test-and-release pipeline — from architecting a CI/CD pipeline mimicking Jenkins to running a production system through staged validation (backtesting, paper trading, live deployment) before it ever touched real capital. Strong in Python and C/C++, comfortable integrating REST APIs, WebSockets, and SQL/NoSQL databases, and experienced tracing hard, intermittent production bugs to their root cause rather than just patching symptoms.

TECHNICAL SKILLS

Languages: C, C++, Python, Java

Frameworks & Libraries: React, Flask, pandas, NumPy

Databases & Version Control: PostgreSQL, MongoDB, SQLite (SQL/NoSQL), Git, GitHub

Testing, CI/CD & Systems: Docker, CI/CD Pipeline Architecture, WebSockets, REST APIs, Real-Time Systems, Root-Cause Debugging

Instrumentation: C#, WPF, NI-VISA, SCPI

PROJECTS

Jenkins-Style CI/CD Pipeline Simulator | *Docker, Python, Webhooks*

- Engineered a distributed CI/CD pipeline simulator explicitly mimicking Jenkins' master-worker architecture, with a central controller handling job scheduling and webhook-triggered builds.
- Built Docker-based worker nodes that execute isolated, containerized builds — hands-on practice with the DevOps and container-orchestration fundamentals behind CI/CD tooling.

Automated Options Trading System — Nifty 50 (India) | *Python, pandas, NumPy, Kite Connect API, WebSockets*

- Designed and deployed an end-to-end options trading system combining VWAP trend signals, IV regime filtering, and futures OI into a weighted confidence score; pushed every change through a staged validation pipeline — backtesting, then paper trading, then live deployment — before it touched real capital.
- Built a production pipeline on a cloud VPS (SEBI 2026-compliant) executing live orders via the Kite Connect API, with WebSocket-based real-time position monitoring, a daily-loss kill switch, and broker-side stop-loss orders as an independent safety net.
- Built the full data and research stack in Python — cleaned ~3 years of tick, minute, and daily market data across 5 sources; validated risk parameters via controlled parameter sweeps rather than heuristics.
- Achieved a 64.9% win rate across 154 backtested trades (2023–2026); caught and corrected a look-ahead bias and an inverted-signal momentum rule through systematic, falsification-driven testing — discarding rules that failed under scrutiny rather than shipping them.

HappyReviews — QR-Based Google Review Collection SaaS | *React, Vite, Supabase, Gemini API, Razorpay API*

- Built and shipped a full-stack SaaS product end-to-end (React/Vite frontend, Supabase for database, auth, and serverless functions, deployed on Vercel) letting businesses collect Google reviews via an in-store QR flow; live with 2 paying customers within the first month.
- Integrated the Gemini API for AI-generated review drafts and OCR-based menu digitization, and built a complete Razorpay subscription billing system with autopay, trial-to-paid conversion, and webhook-driven state reconciliation.
- Diagnosed and fixed a production race condition causing silent review-submission failures on mobile — systematically ruled out RLS, deployment, and caching causes before isolating an unawaited async database write being interrupted by tab backgrounding.

Bench Test Automation Suite | *C#, WPF, Python, Flask, NI-VISA, SCPI, SQLite*

- Simulated a Keithley-style DMM/DC power supply using pyvisa-sim in Python, wiring up standard SCPI commands (*IDN?, MEAS:VOLT:DC?, OUTP ON/OFF) behind a Flask API that the C# client calls over HTTP.
- Designed the test sequencer around an ITestStep interface in C#, where each step's Execute() method returns a TestResult carrying expected vs. actual values — built for polymorphic test-step types rather than one hardcoded script.
- Built a WPF UI with a data-bound DataGrid that updates in real time as each test step completes, showing live pass/fail status.
- Debugged and fixed a CORS and JSON-serialization issue that broke the first end-to-end run between the C# client and Flask API, tracing it to missing headers and improperly serialized state.

EDUCATION

B.E. in Computer Science and Business Systems, BMS College of Engineering | Bengaluru, India | Jul 2027

CGPA: 7.5

Pre-University, Excellent PU College | Mangaluru, India | Mar 2023

Percentage: 87.2%

CERTIFICATIONS

Google Generative AI Certification (Google Cloud Skills Boost) • Oracle Database Programming (Oracle Academy) • MATLAB for Plots & Statistics (MathWorks) • Nutanix Certified Associate – Multicloud Infrastructure

LEADERSHIP & ACTIVITIES

- Coordinator — PhaseShift '24, Utsav '25, and Hospitality (BMSCE): led campus decor teams and hospitality operations for major college-wide events. Volunteer, NSS — plantation drives, Mental Health Day, and community outreach campaigns.